

assignment.py

```
1  #input student's roll , marks, float
2  s_name=input("Enter the student name: ")
3  roll=int(input("Enter the student roll number: "))
4  phy=float(input("Marks for physics: "))
5  chem=float(input("Marks for chemistry: "))
6  bio=float(input("Marks for biology: "))
7
8  details={
9      "name": s_name,
10     "Roll number: ": roll,
11     "Physics: ": phy,
12     "Chemistry": chem,
13     "Biology": bio}
14
15
16  total=phy+chem+bio
17  print("Total marks: ", total)
18
19  percentage=total/300*100
20  print("Percentage: ", percentage)
21
22  # for key, value in details.items():
23  #     print(key, value)
24
25
26
27  status="pass" if (phy>=40 and chem>=40 and bio>=40) else "fail"
28  print("Result: ", status)
29
30  #pass fail if each marks>=40
31  # if phy>=40 and chem>=40 and bio>=40:
32  #     print("Result: Pass")
33  # else:
34  #     print("Result: Fail")
35
36
37
38  if percentage>=90:
39      print("Grade: A+")
40  elif percentage>=80 and percentage<90:
41      print("Grade: A")
42  elif percentage>=70 and percentage<80:
43      print("Grade: B+")
44  elif percentage>=60 and percentage<70:
45      print("Grade: B")
46  elif percentage>=50 and percentage<60:
47      print("Grade: C+")
48  elif percentage>=40 and percentage<50:
49      print("Grade: C")
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50     else:
51         print("Grade: F")
52
53     if status == "pass":
54         if percentage >= 80:
55             print("Distinction")
56         elif percentage >= 60 and percentage < 80:
57             print("First class")
58         elif percentage >= 50 and percentage < 60:
59             print("Second class")
60         elif percentage >= 40 and percentage < 50:
61             print("Third class")
62
63     #Nested loop and in (membership test) operator
64     #alternatives
65
66     if percentage >= 90:
67         grade = "A+"
68     elif percentage >= 80:
69         grade = "A"
70     elif percentage >= 70:
71         grade = "B+"
72     elif percentage >= 60:
73         grade = "B"
74     elif percentage >= 50:
75         grade = "C+"
76     elif percentage >= 40:
77         grade = "C"
78     else:
79         grade = "F"
80
81     print("Grade:", grade)
82
83     if status == "pass":          # Outer if
84         if grade in {"A+", "A"}: # Nested if + set membership test
85             print("Distinction")
86         elif grade in {"B+", "B"}:
87             print("First Class")
88         elif grade == "C+":
89             print("Second Class")
90         else:
91             print("Third Class")
92
93     status = "pass " if (phy >= 40 and chem >= 40 and bio >= 40) else "fail"
94
95     print("Student Report Card".center(40, "-"))
96     print(f"Name: {s_name}")
97     print(f"Roll Number: {roll}")
98     for key, value in details.items():
99         print(f"{key}: {value}")
100    print(f"Total Marks: {total}")
101    print(f"Percentage: {percentage:.2f}%")
102    print(f"Grade: {grade}")

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103     print(f"Result: {status}")
```